

Data Governance

Dimitrios Sargiotis

Data Governance

A Guide

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To my beloved family:

To Eirini, my wonderful wife, for her endless support and love.

To Konstantinos, my cherished son, who inspires me every day.

To my father, who left us in November 2017, for his enduring guidance, strength, and the ethos of integrity and perseverance that he instilled in me. Your memory and values continue to guide me every day.

To Paraskevi, my dear mother, for her unwavering belief in me.

To Theodora, my supportive sister, for always being there.

And to the new generation, may you navigate and shape the future with wisdom and integrity.

Preface

In the digital era, where data has become the new oil, the importance of managing this invaluable asset cannot be overstated. “Data Governance: A Guide” is a comprehensive resource designed to provide insights, methodologies, and best practices in the field of data governance.

My journey into data governance began over a decade ago, sparked by the challenges and opportunities I observed in the rapidly evolving data landscape. The explosion of data in the digital age has not only transformed how organizations operate but also raised complex issues surrounding data security, privacy, and quality. This book is a culmination of years of research, practical experience, and discussions with experts in the field.

The primary aim of this guide is to demystify data governance and make it accessible to professionals across various sectors. Whether you are a data management veteran or new to the field, this book offers valuable insights into establishing a robust data governance framework.

Throughout the chapters, I delve into the core principles of data governance, addressing key components such as data quality, security, privacy, and management. Real-world case studies are interspersed to illustrate the tangible benefits and challenges of implementing data governance in diverse organizational contexts.

This book also explores the future landscape of data governance, considering emerging trends and technologies such as artificial intelligence, machine learning, and blockchain. The goal is to equip readers with the knowledge to not only navigate the current landscape but also to anticipate and prepare for future developments.

Authoring this book has been a journey of discovery and reaffirmation of the critical role data governance plays in modern organizations. It is my sincere hope that this guide will serve as a valuable resource for you, sparking innovative ideas, strategies, and a deeper understanding of data governance.

Welcome to the journey of mastering data governance.

Marousi, Greece

Dimitrios Sargiotis

Acknowledgments

The journey of creating “Data Governance: A Guide” has been both enriching and challenging. This book, a labor of love and dedication, owes its existence to the collective efforts of many remarkable individuals whose support, expertise, and encouragement have been invaluable.

First and foremost, I extend my heartfelt gratitude to the myriad of data governance professionals and academics whose pioneering work and insightful discussions have laid the foundation for this guide. Their dedication to advancing our understanding of data governance has been a source of constant inspiration and learning.

I am particularly grateful to my peers and colleagues in the data management community. Their willingness to share experiences, challenges, and success stories has enriched this guide with a diversity of perspectives that reflect the real-world complexities of data governance.

To my family, who has stood by me with patience and understanding through the countless hours dedicated to writing and research, I owe a debt of gratitude. Your encouragement and belief in the value of this work have been my greatest motivators.

I also wish to acknowledge the invaluable contributions of my editorial team, whose expertise and attention to detail have greatly enhanced the quality of this guide. Their dedication to excellence has been a key factor in bringing this project to fruition.

To the reviewers who generously contributed their time and expertise to provide feedback, thank you. Your constructive critiques have been essential in refining the content and ensuring its relevance and accuracy.

Lastly, to the readers and future data governance practitioners, this guide is for you. It is my hope that it will serve as a valuable resource as you navigate the complexities of data governance in your professional journey.

The completion of this guide marks not an end, but a beginning—a step toward a future where data governance is recognized not just as a necessity, but as a cornerstone of ethical and effective data management.

Competing Interests

In alignment with Springer Nature’s commitment to transparency and integrity, I, Dimitrios Sargiotis, wish to disclose any competing interests—both financial and non-financial—that might have influenced the content of “Data Governance: A Guide.”

Financial Interests

- As the sole author of this guide, I have not received direct financial compensation specifically for the creation of this book. My professional affiliations, including any consultancy roles, research funding, or associations with entities that have a vested interest in data governance, are disclosed herein.
- I declare that there are no royalties or financial benefits received from third parties directly for the writing and publication of this book. Any future royalties or financial benefits will stem directly from the sales of the book itself, without external endorsements or funding.

Non-financial Interests

- My commitment to advancing the field of data governance is both a professional and personal interest. I hold no advisory board positions, consultancy roles, or affiliations with organizations that might present a conflict of interest regarding the book’s content.
- I am dedicated to providing an objective, balanced view on data governance, free from personal beliefs, affiliations, or relationships that could be perceived to bias the content of this guide.

This declaration serves to assure the reader of my commitment to ethical and transparent publishing. My aim in authoring “Data Governance: A Guide” is to furnish the reader with unbiased, practical insights into data governance, unswayed by competing interests.

In the spirit of full disclosure, should any changes occur in my affiliations or financial interests after the publication of this book, I commit to updating this statement in subsequent editions to reflect such changes accurately.

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Abbreviations

AI	Artificial Intelligence
CCPA	California Consumer Privacy Act
DLP	Data Loss Prevention
ETL	Extract, Transform, Load
GDPR	General Data Protection Regulation
HIPAA	Health Insurance Portability and Accountability Act
IAM	Identity and Access Management
IoT	Internet of Things
ISO	International Organization for Standardization (e.g., ISO 27001)
MDM	Master Data Management
ML	Machine Learning
PCI DSS	Payment Card Industry Data Security Standard
RBAC	Role-Based Access Control
SOX	Sarbanes-Oxley Act
SQL	Structured Query Language

Glossary

Data governance	The overall management of the availability, usability, integrity, and security of data used in an organization
Data integration	The process of combining data from diverse sources into a single, unified view
Data lifecycle management	The process of managing the flow of data through its lifecycle from creation and initial storage to the time when it becomes obsolete and is deleted
Data privacy	Ensuring that personal information is handled, stored, and used in compliance with privacy laws and standards
Data quality	The measure of data's condition, focusing on accuracy, completeness, reliability, and relevance
Data security	Protecting data from unauthorized access and corruption throughout its lifecycle
GDPR compliance	Adherence to the General Data Protection Regulation, a regulation in EU law on data protection and privacy
Master data management (MDM)	A method that defines and manages the critical data of an organization to provide, with data integration, a single point of reference
Metadata management	The administration of data that describes other data, providing context and aiding in data discovery
Regulatory compliance	Adhering to laws, regulations, guidelines, and specifications relevant to business processes

Annotations

Data privacy	Data privacy concerns the proper handling, processing, and storage of personal information. Organizations must ensure they comply with legal standards and ethical considerations to protect individual privacy rights
Data quality	Data quality is fundamental to reliable decision-making. It involves ensuring accuracy, completeness, and reliability of data. Poor data quality can lead to erroneous conclusions and decisions, impacting organizational performance
Data security	Data security involves protecting data from unauthorized access and breaches. It is a critical aspect of data governance, especially in an era where cyber threats are increasingly sophisticated
Master data management	Master Data Management (MDM) is a method of managing the organization's critical data. It provides a single point of reference to ensure that the organization's data is consistent, accurate, and controlled
Metadata management	Metadata management is the administration of data that describes other data. It is essential for understanding data assets in an organization and plays a key role in data governance by providing context and aiding in data discovery
Regulatory compliance	Compliance with regulations such as GDPR (General Data Protection Regulation) is crucial for any organization handling personal data. Non-compliance can result in hefty fines and damage to reputation

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